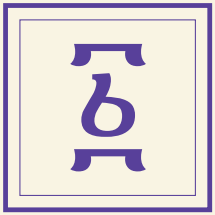
Finding the way, step by step

Weekend 1: Large language models

Carlos Cotrini
AI in Industry, ETH Zurich
Weekend 1, October 2026



Section 1

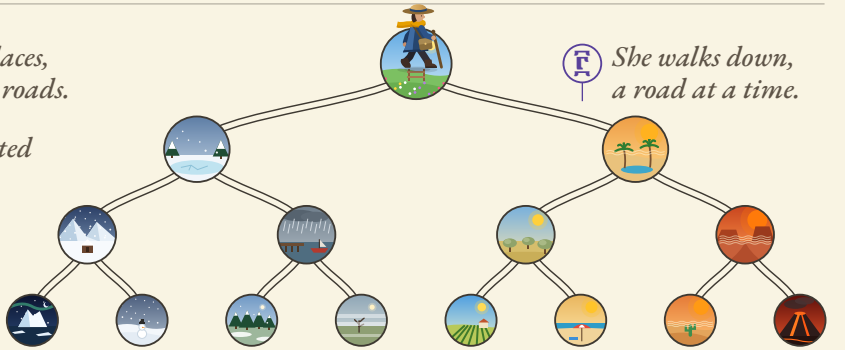
The village

Mara carries letters down the valley

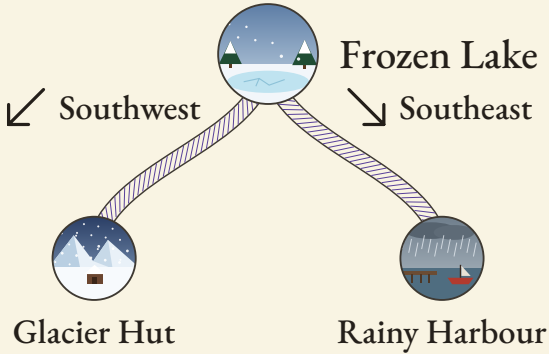
Fifteen places, joined by roads.

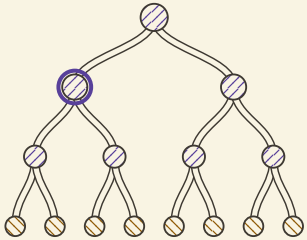
An invented village.

She walks down, a road at a time.

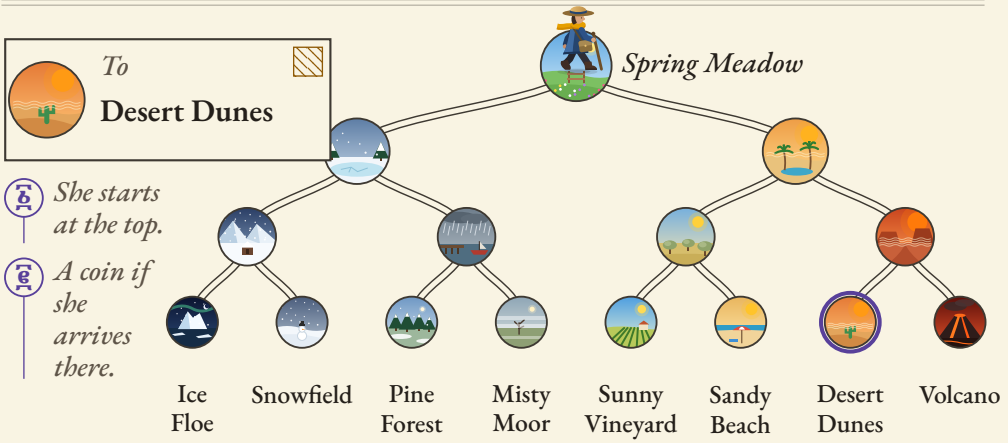


Every place on the way has two roads

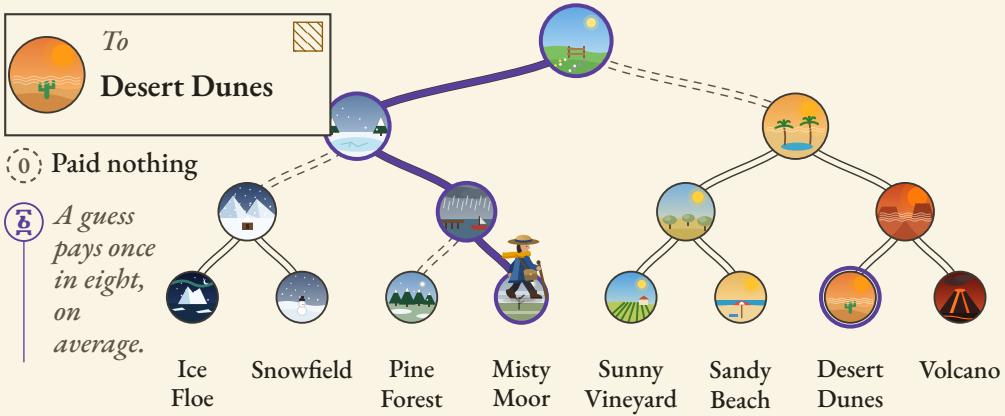




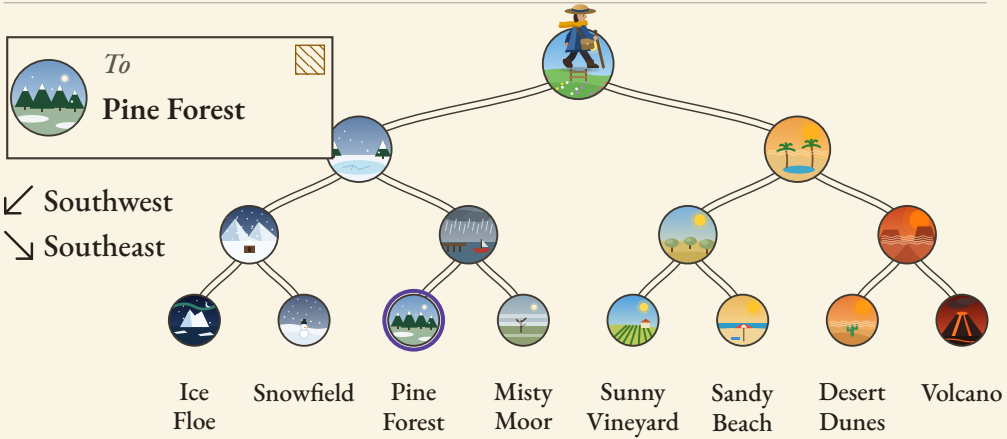
Each letter names a place in the bottom row

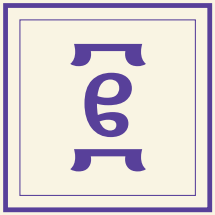


At first she can only guess



Your turn: which road first?





Section 2

The clue

After many letters, a pattern

Letter to	Roads			
Ice Floe	↙	↙	↙	
Snowfield	↙	↙	↘	
Pine Forest	↙	↘	↙	
Misty Moor	↙	↘	↘	
Sunny Vineyard	↘	↙	↙	
Sandy Beach	↘	↙	↘	
Desert Dunes	↘	↘	↙	
Volcano	↘	↘	↘	

She starts at

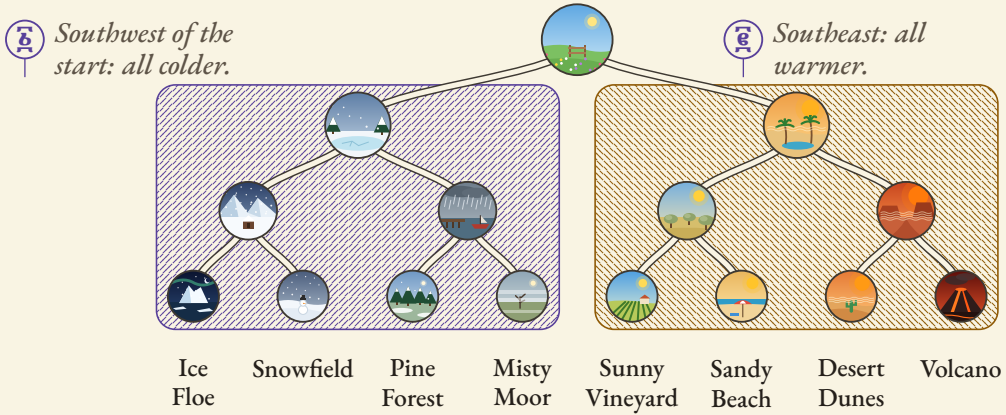


Spring Meadow

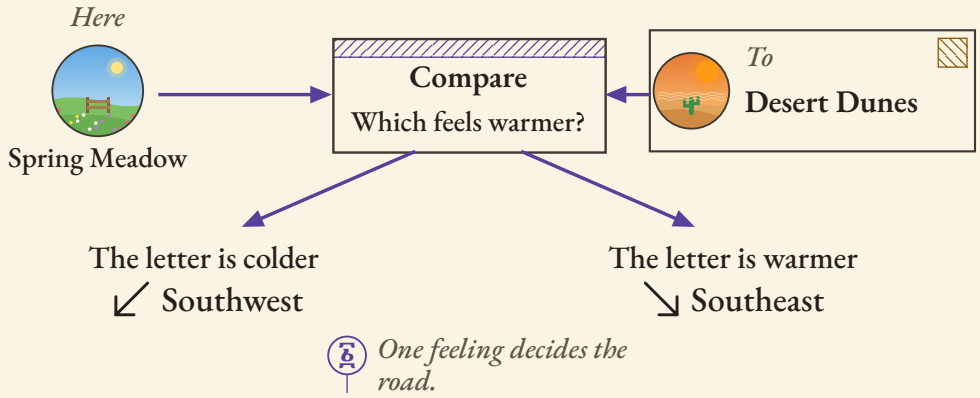
Colder than the start: southwest first.

Warmer than the start: southeast first.

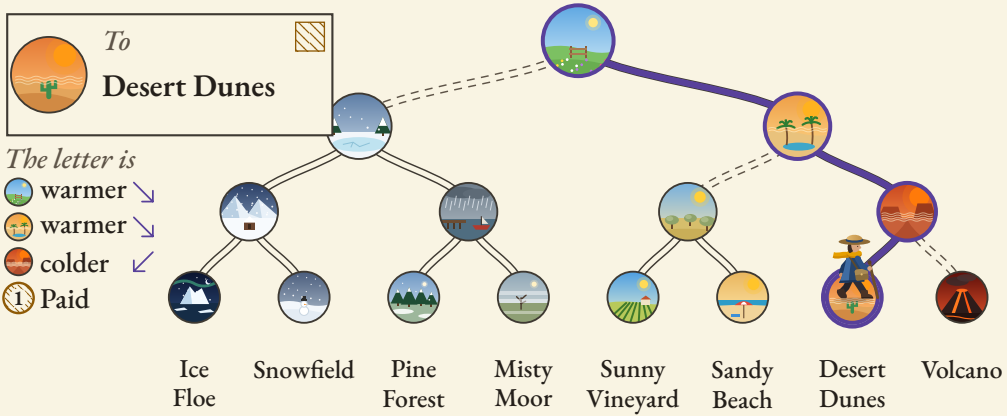
Southwest feels colder, southeast feels warmer



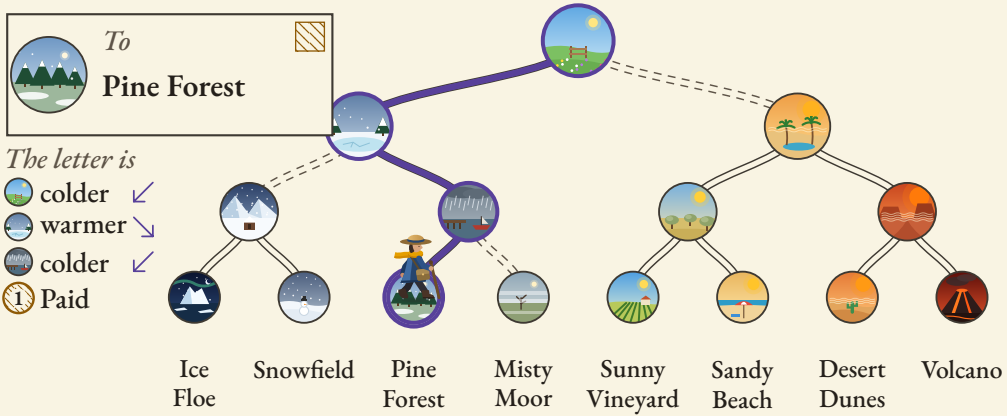
The rule: compare here with the letter



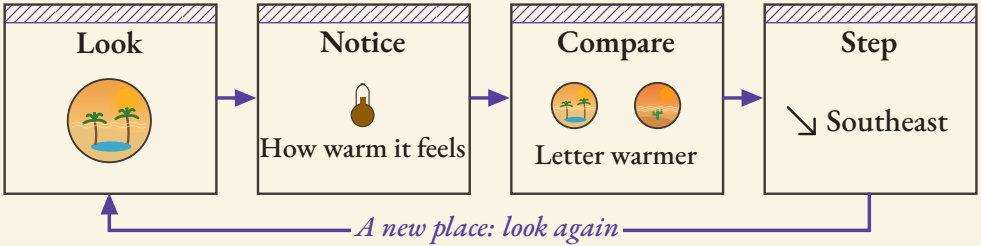
The rule finds Desert Dunes



Your turn again: Pine Forest

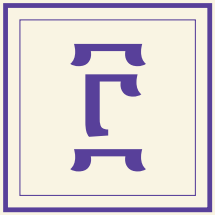


Look, notice, compare, step



She ignores the rest of the picture.

Nobody told her what to notice. The coins taught her.



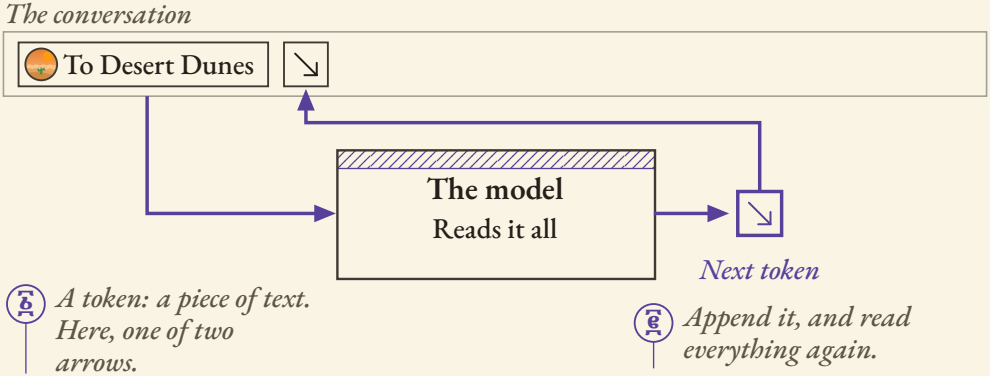
Section 3

Reasoning

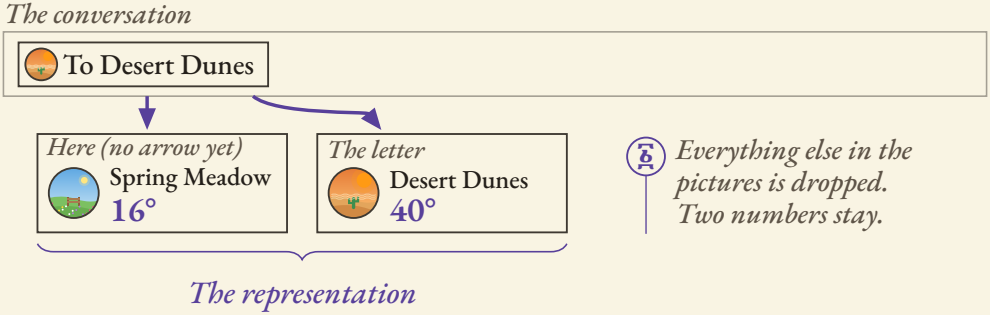
A reasoning model runs the same loop

	Mara	A reasoning model
Look	Where she stands	The question, and its notes so far
Notice	How warm it feels	What matters in the notes
Step	A road	One small step, written down
Repeat	Until she arrives	Until it has an answer
Learn	From the coins	From checking the answers

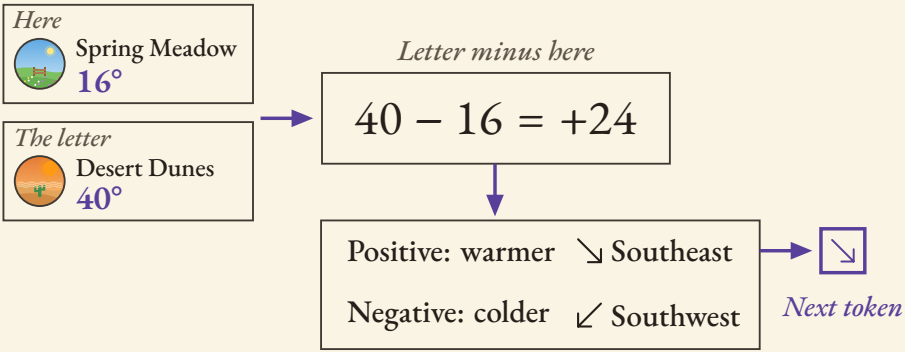
The model writes one token at a time



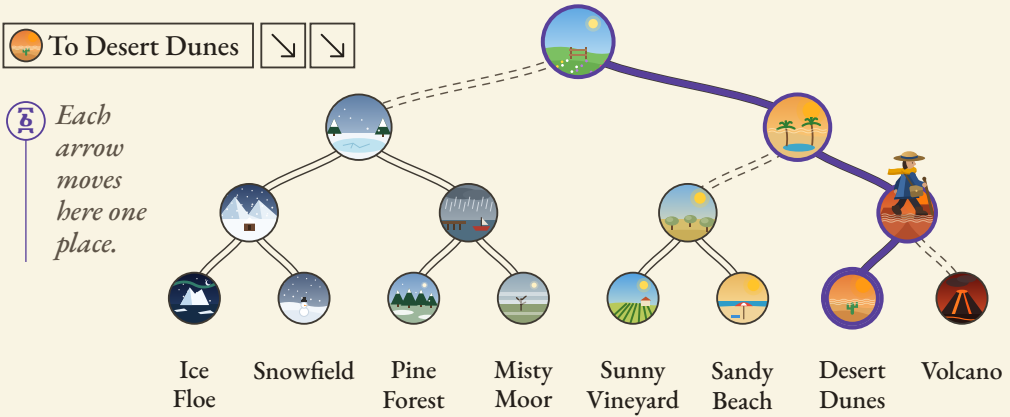
From the conversation, two numbers



A difference picks the next token



Here is worked out from the tokens



Append, and compute again

The conversation	Here	The letter	Difference	Next
To Desert Dunes	16°	40°	+24	↘
To Desert Dunes ↘	34°	40°	+6	↘
To Desert Dunes ↘ ↘	45°	40°	−5	↙
To Desert Dunes ↘ ↘ ↙	Answer: Desert Dunes			1

Each round reads the whole conversation again.

The arrows before the answer are the thinking.

In practice: notes before the answer

Question. A client orders 3 crates at 40 francs each. Orders over 100 francs get 10 percent off. What does the client pay?

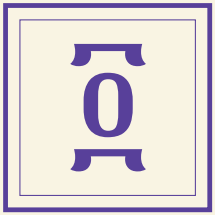
Thinking

3 crates at 40 francs: 120 francs.
120 is over 100, so the discount applies.
10 percent of 120 is 12.
120 minus 12 is 108.

Answer: the client pays 108 francs.

Each line reads the notes so far and adds one small step.

Like Mara: look, notice what matters, step.



Section 4

Training

The model keeps a table of chances

Here	The letter	Southwest	Southeast
Spring Meadow	Ice Floe	21%	79%
Spring Meadow	Desert Dunes	67%	33%
Frozen Lake	Pine Forest	14%	86%
Palm Oasis	Sunny Vineyard	75%	25%
Red Canyon	Desert Dunes	86%	14%
Rainy Harbour	Pine Forest	81%	19%

And 18 more rows



A row for each place and letter: twenty-four rows.

At first, random.

One iteration: pick a row at random

Here	The letter	Southwest	Southeast
Spring Meadow	Ice Floe	21%	79%
Spring Meadow	Desert Dunes	67%	33%
Frozen Lake	Pine Forest	14%	86%
Palm Oasis	Sunny Vineyard	75%	25%
Red Canyon	Desert Dunes	86%	14%
Rainy Harbour	Pine Forest	81%	19%

And 18 more rows



Picked at random.

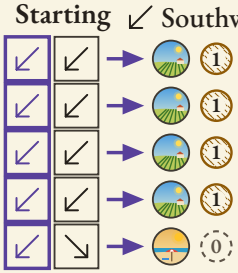
Step two: send walks, half each way

Here

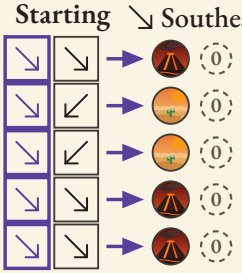
Palm Oasis

To Sunny Vineyard

Later roads: drawn from the table.

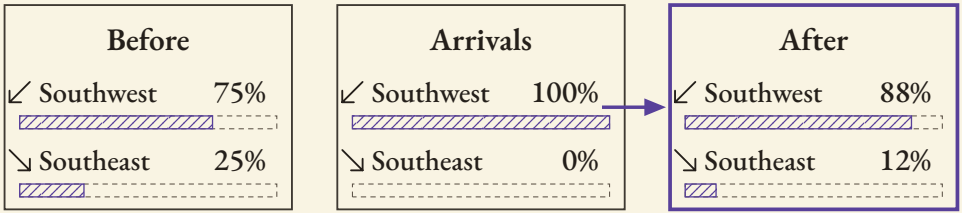


4 of 5 arrived



0 of 5 arrived

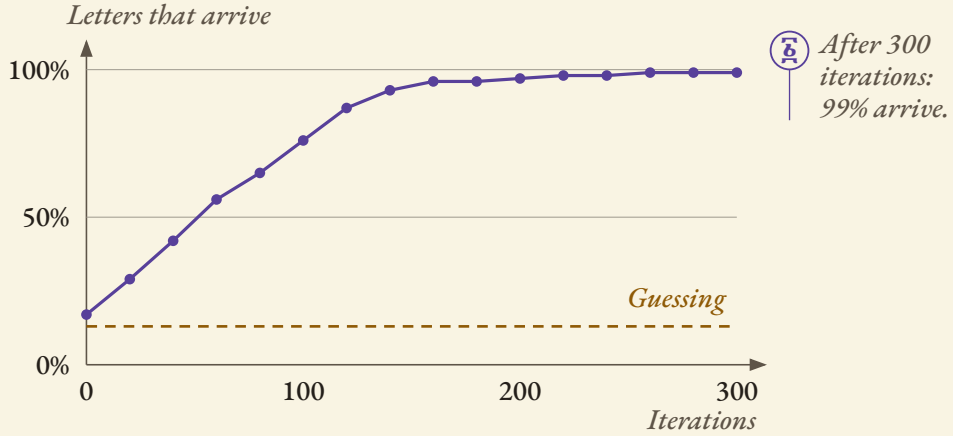
Step three: move the row toward what arrived



4 walks arrived

After: halfway from before to the arrivals.

Repeat, many times



After training, the table knows the rule

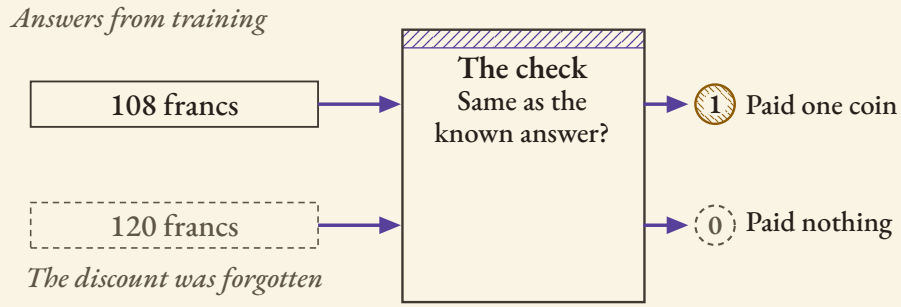
Here	The letter	Southwest	Southeast
Spring Meadow	Ice Floe	100%	0%
Spring Meadow	Desert Dunes	1%	99%
Frozen Lake	Pine Forest	1%	99%
Palm Oasis	Sunny Vineyard	100%	0%
Red Canyon	Desert Dunes	100%	0%
Rainy Harbour	Pine Forest	100%	0%

And 18 more rows

Colder letter: southwest. Warmer: southeast.

Learned from the coins alone.

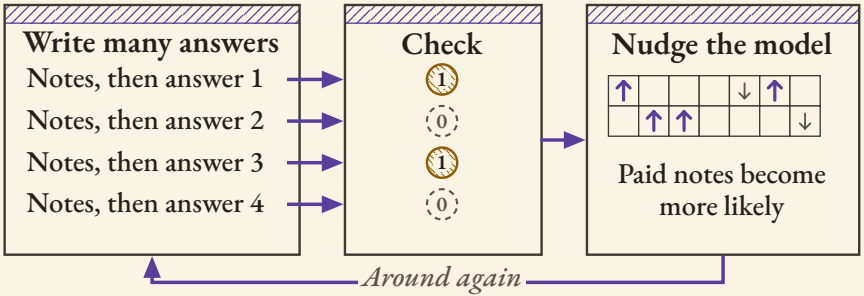
A check pays the answer



The check reads the answer only. Nobody grades the notes.

Training uses questions whose answer is known.

A model learns its notes from the check



Mara learned the clue from her coins.

A model learns its notes from the check.

DeepSeek-R1 learned to reason from a check

- Its first model, R1-Zero, learned by trial and error alone. Nobody wrote reasoning for it.
- It was paid only when the final answer of a maths or coding problem was right, and neatly written.
- Reasoning appeared by itself: longer notes, checking its own work, trying another way.
- The released DeepSeek-R1 adds some written examples and more training.

https://doi.org/10.1038/s41586-025-09422-z

D. Guo et al. (DeepSeek-AI), 2025. *DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning*. *Nature* 645, 633-638.

The idea behind most reasoning models

- Look at where you stand
- Notice what matters
- Take one small step, and write it down
- Repeat until there is an answer
- Learn from a check of the answer which steps pay off

Example: D. Guo et al., *Nature* 645, 633-638 (2025), doi:10.1038/s41586-025-09422-z.